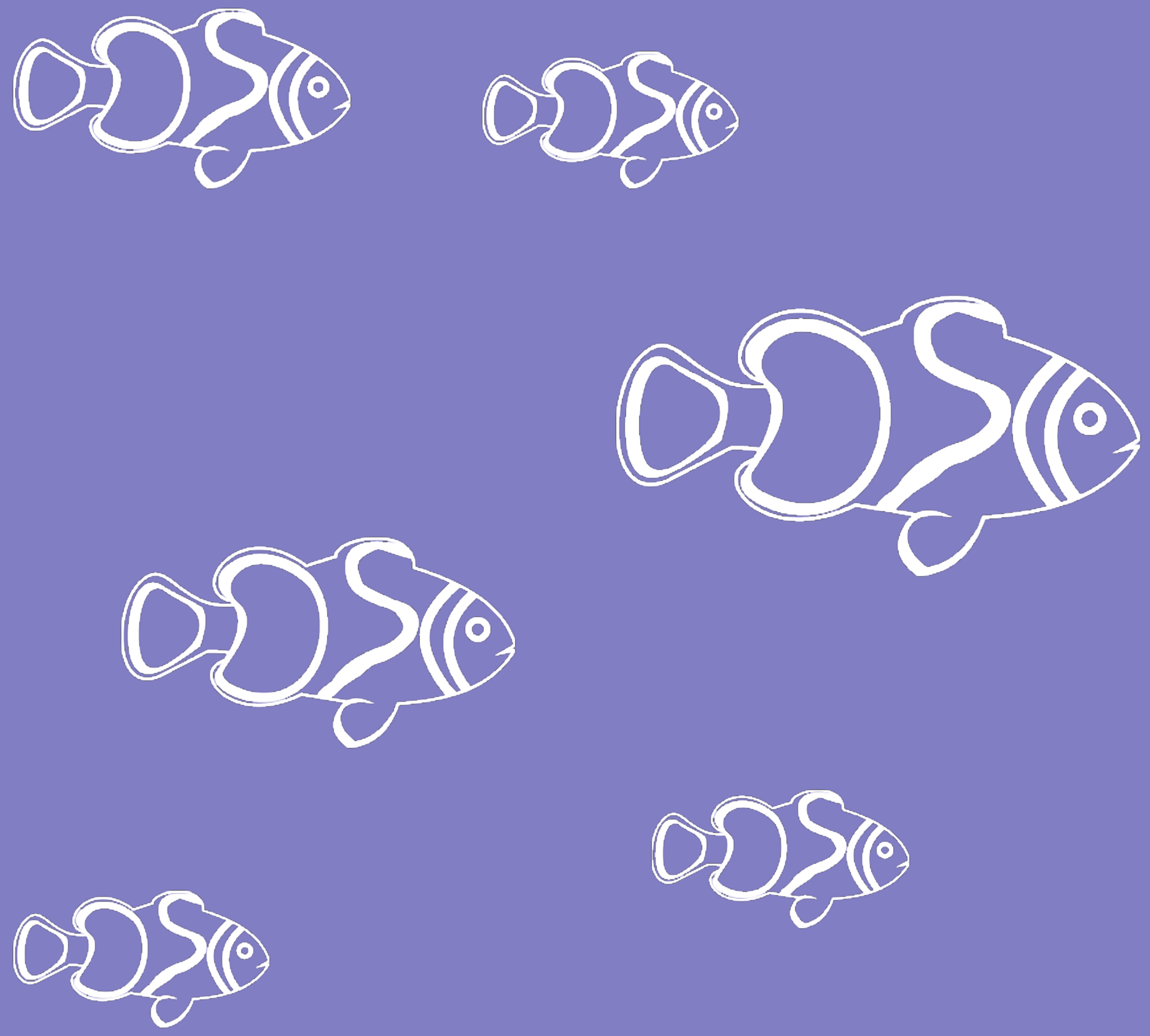
Complex Adaptive Systems

A Short Guide



Overview

Complex adaptive systems are all around us from financial markets to ecosystems to the human immune system and even civilization itself. These systems consist of many agents that are acting and reacting to each other's behavior, out of this often chaotic set of interactions emerges global patterns of organization in a dynamic world of constant change and evolution where nothing is fixed.

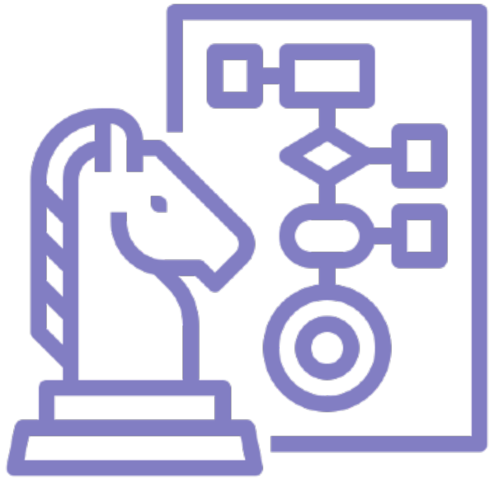
In this guide we by talking about adaptation itself, where we will be discussing cybernetics and looking at how systems regulate themselves to respond to change. We'll go on to talk about the dynamics of cooperation and competition with game theory.

We looking at the process of evolution as a powerful and relentless force that shapes complex adaptive systems on the macro-level. Finally the ideas of resilience is introduced here and contrasted with robustness.

Guide Contents

1. Adaptive System

How systems adapt through regulatory processes and respond to their environment.



2. Game Theory

We look at situations of cooperation & competition between adaptive agents

3. Evolution

How complex adaptive systems manage to adapt to a changing environment

4. Resilience

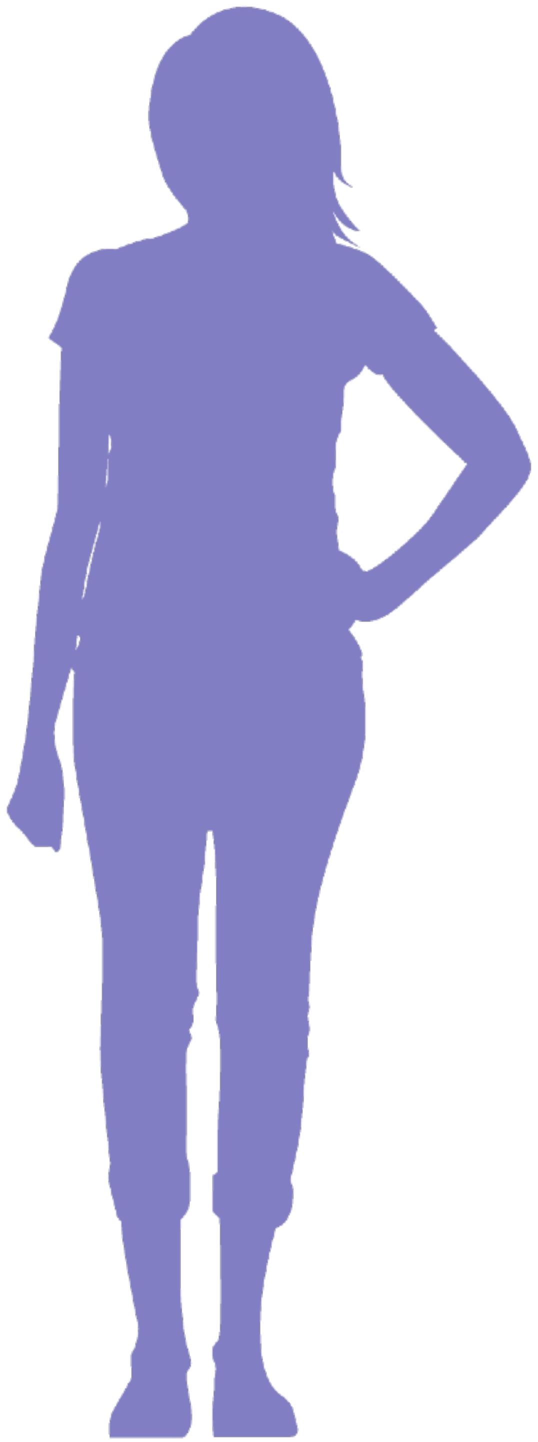
Learn the difference between robustness and resilience

Adaptive Systems

What is an Adaptive System?

We can define adaptation as the capacity for a system to change its state in response to some change within its environment. An adaptive system is then, a system that can change given some external perturbation, and this is done in order to optimize or maintain its condition within an environment by modifying its state.
Adaptation simply means the system can generate a number of different responses to a set of changes within the state of its environment. This adaptive capacity gives a system some flexibility that improves its performance.

The growth of a plant or fungus towards a source of light, what is called phototropism, is an example of adaptation. It is designed to respond to external influence that creates a response within the system. Entities that are capable of more advanced forms of adaptation have specialized subsystems dedicated to regulating this process of adaptation. We call these specialized components regulatory or control systems.



Agency

All adaptive systems regulate some process and they do this in order to maintain and develop their structure and functioning. For example, plants process light and other nutrients and their adaptive capacity enables them to alter their state so as to intercept more of those resources. The same is true for bacteria and animals, the same is true for a basketball team or a business, they all have some conception of value that represents whatever is the resource that they require, whether that is sunlight, fuel, food, money, etc.

This creates what we can call a value system, that is to say, whatever structure or process they are trying to develop forms the basis for their conception of value. Agents use their agency to act and make choices in the world to improve their status with respect to whatever it is they value; this value system may be very simple or very complex. All adaptive systems have a degree of agency. Agency is the capacity to make choices based upon information and act upon those choices autonomously to affect the state of their environment.

Cybernetics

Cybernetics is the study of control, communications and information processing within systems of all kinds, biological, mechanical and social. Norbert Wiener - one of the founders of the subject - defined cybernetics as “the scientific study of control and communication in the animal and the machine.” The word cybernetics comes from Greek word meaning “governance” or “to steer, navigate or govern.”

The primary object of study within cybernetics are control systems that are regulated by negative feedback loops. There are essentially just three components to any given control mechanism. Firstly, there needs to be some form of a sensor for feeding information into the system. Secondly, there needs to be a controller that contains the logic or set of instructions for processing this information. Lastly, an actuator that executes some action in order to affect the state of the system or its environment.

Homeostasis

Many types of systems require both a continuous input of resources from their environment and the capacity to export entropy back to the environment in order to maintain their given level of functionality. A business organization is one example, requiring a continuous revenue stream to pay its employees and suppliers, while also producing a certain amount of waste material that it must externalize.

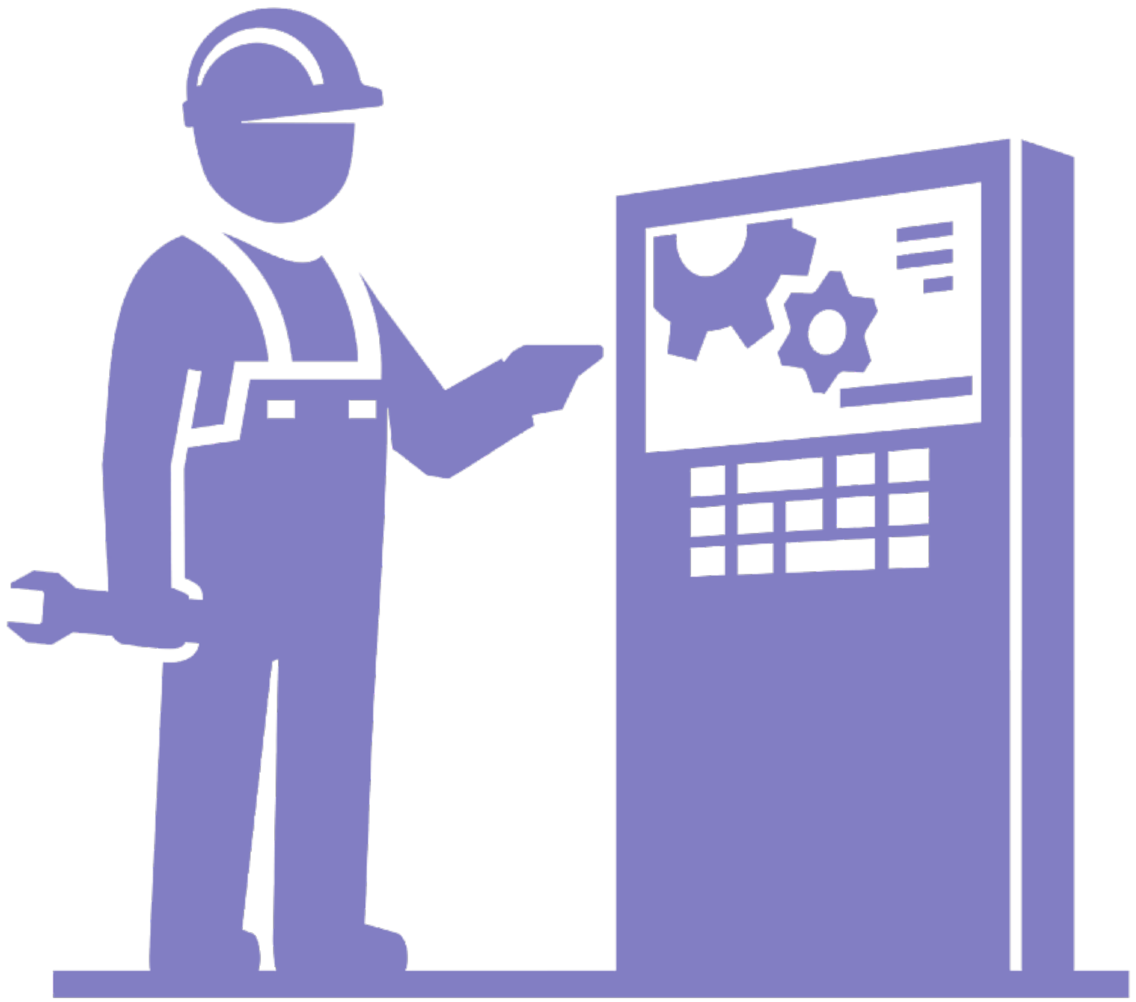
In order for these systems to maintain their intended level of functionality - what we might call their normal or equilibrium state - they must have an environment that is conducive to providing them with these required conditions. Within ecology and biology the term homeostasis is used to describe this phenomenon.

The word Homeostasis derives from the Greek word meaning homos or "similar" and stasis meaning "standing still". It is the state of a system in which variables are regulated so that internal conditions remain stable and relatively constant, despite changes within the system's environment.

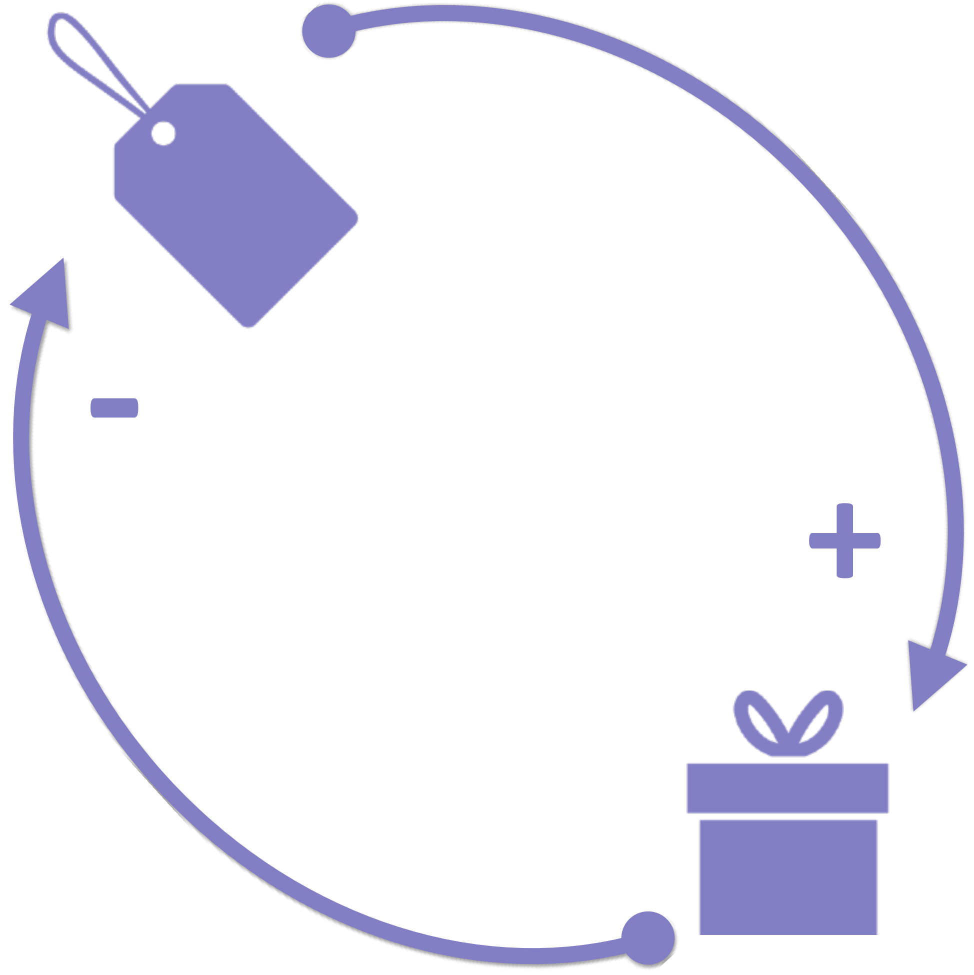
Regulation

In order for systems to maintain homeostasis, there needs to be some kind of regulatory mechanism; what we also call a control system. This control mechanism has to regulate both the system's internal and external environment to ensure that the environmental conditions are within the given set of parameters that will enable the internal processes of the system to function at a normal, equilibrium state.

Cybernetics is the area of systems theory that studies these regulatory mechanisms. Cybernetics comes from a Greek word which means to steer or guide, and this is exactly what a control system is designed to do. It is designed to guide the system in the direction of the set of environmental parameters that are best suited for it to maintain homeostasis. Systems use their regulatory mechanisms to maintain their internal composition within just the right parameters required for the needed processes to take place.



Feedback Regulation

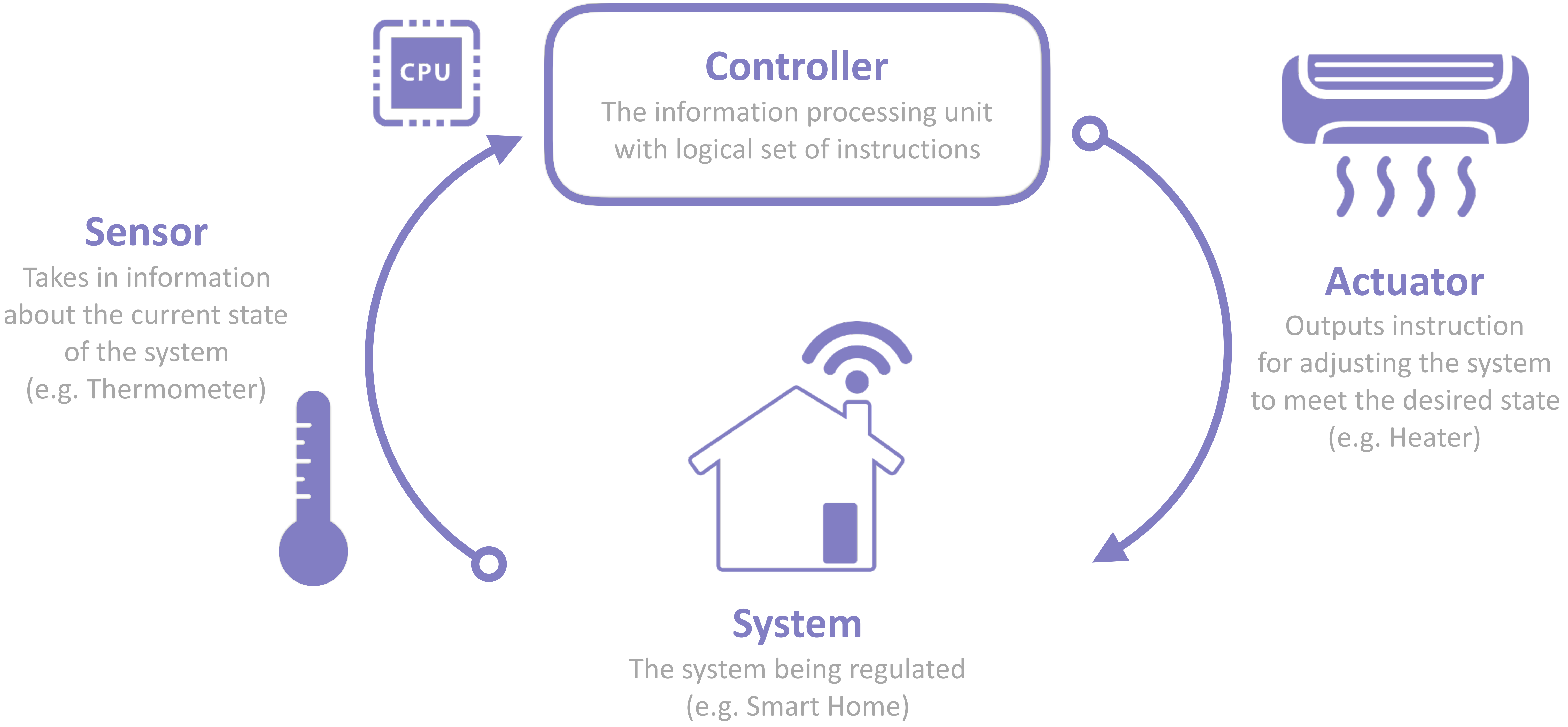


Feedback loops are a fundamental object of study within cybernetics in that they are accountable for the process of regulation within all control systems. Negative feedback is at the heart of processes of regulation and control of all kind.

An example of this might be the feedback loops that regulate the temperature of the human body. Different body organs work to maintain a constant temperature within the body by either conserving or releasing more heat. Through sweating and capillary dilation, they counter-balance the fluctuations in the external environment's temperature.

Another example of a negative feedback control process would be between the supply and demand of a product. The more the supply of a product the lower the price will become which would feedback to induce the producers to lower production. This is a decentralized form of regulatory process, without any one person directing the market the feedback prevents overpricing and overproduction.

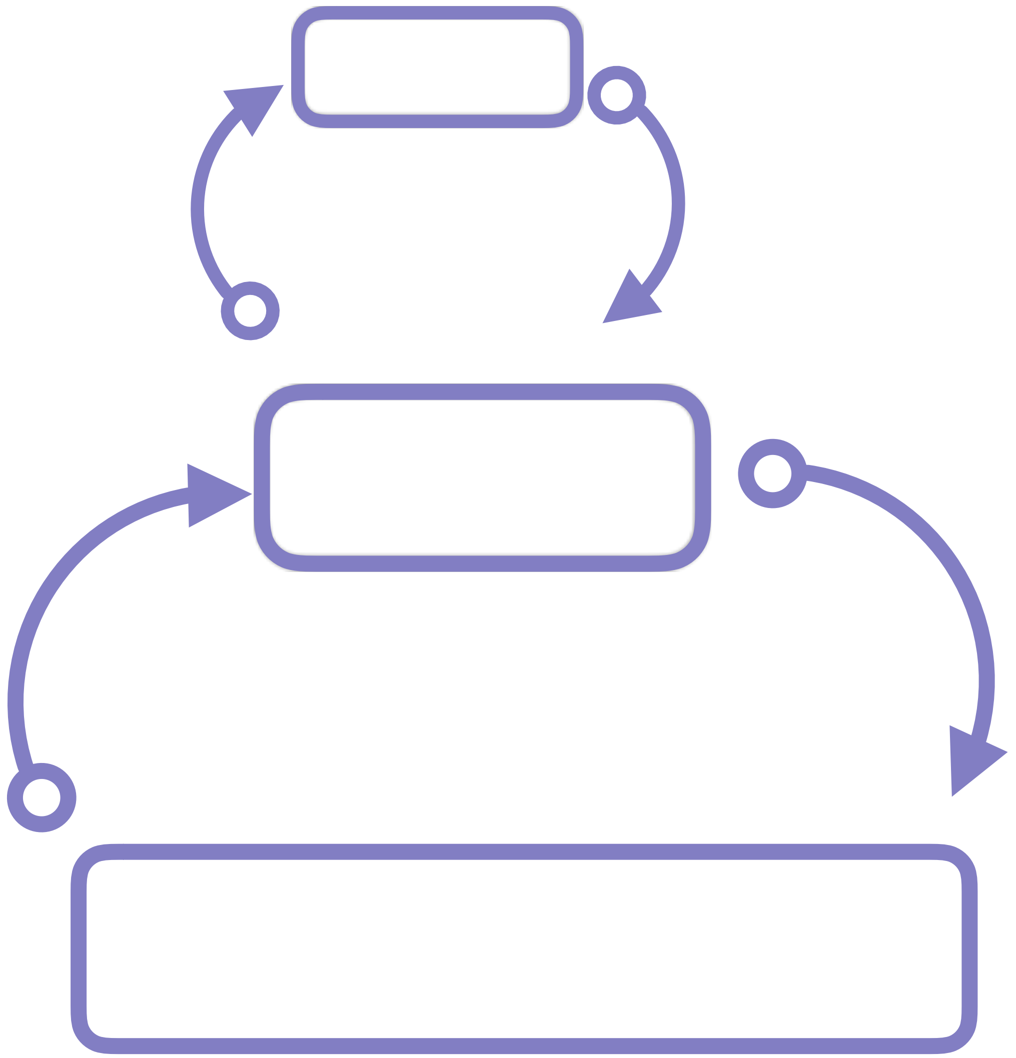
How It Works: Control Systems



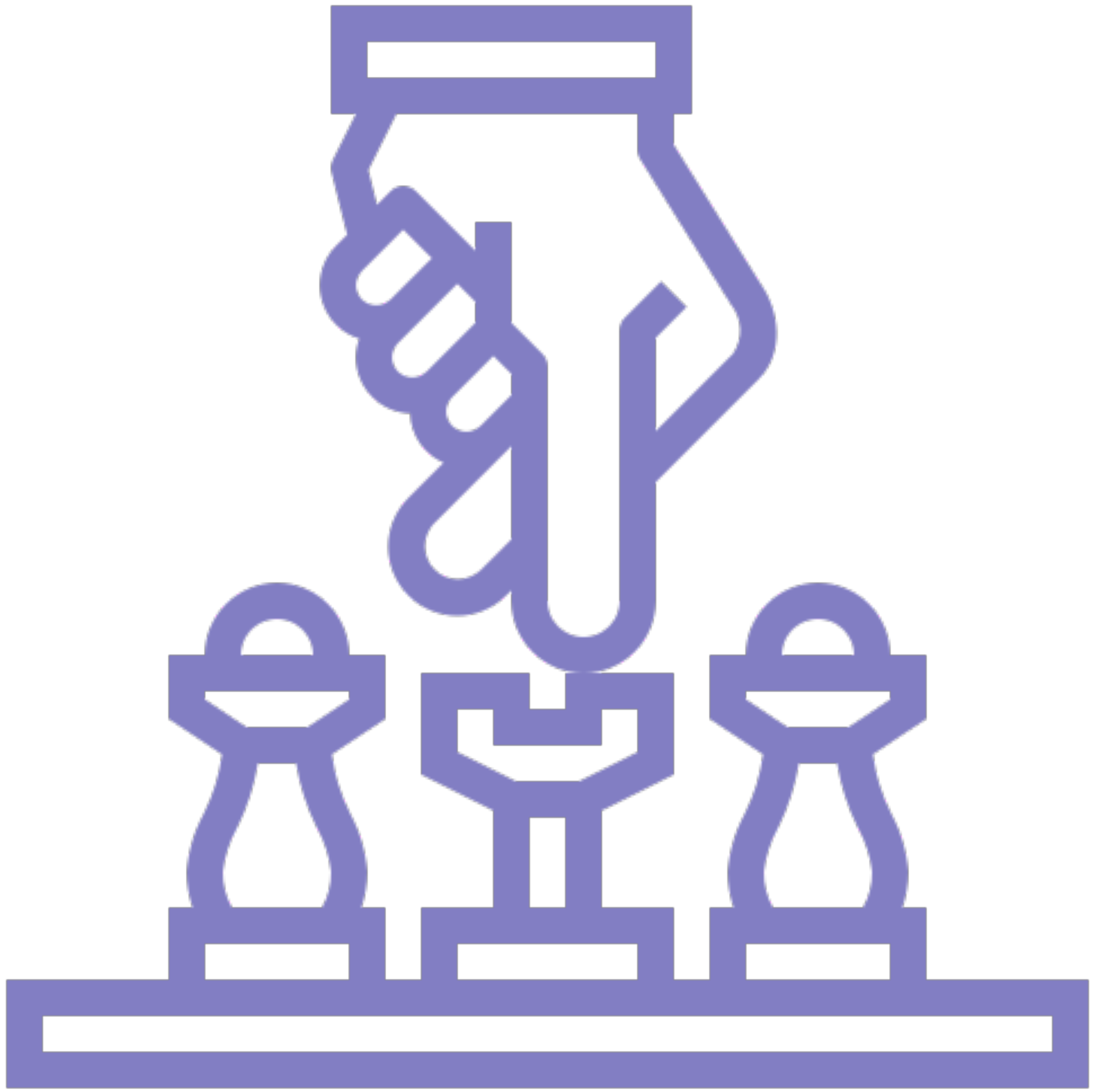
Second Order Cybernetics

New cybernetics or second-order cybernetics is sometimes described as the "cybernetics of cybernetics". It investigates the construction of models of cybernetic systems looking beyond the issues of the “first order” cybernetics and the emphasis on control, recognizing that the investigator is also part of the system, and of the importance of autonomy, self-referentiality, and self-organizing capabilities of complex systems.

This is a recognition that the investigators of a system can never see how it works by standing outside it because the investigators are always engaged themselves in a cybernetic process of observation with the system being observed; that is, when investigators observe a system, necessarily they affect it and are affected by it. Thus second order cybernetics introduces the observers conceptual system as a “second oder” to the regulatory process, through which it can be adjusted, adapted and we can get such phenomena as learning.



Game Theory

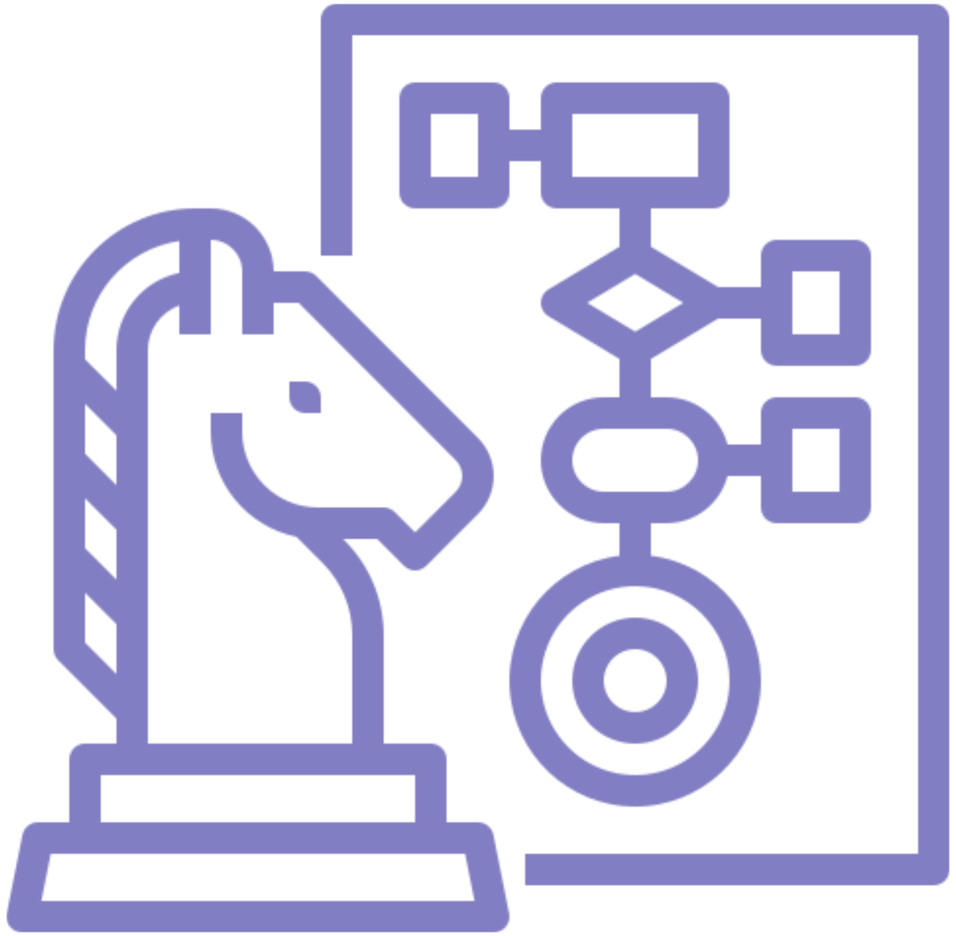


Game Theory

When two adaptive systems interact there comes to form a dynamic of cooperation and competition as the pursuit of the valued ends of one agent becomes interdependent with that of others. Game theory is the study of these interactions between adaptive agents. Game theory studies the strategic interaction between agents engaged in relations of cooperation and conflict.

Game theory is used in many different areas whenever there are agents that act according to some form of logic to maximize their pay-off within some rule-bound interaction, such as politics, economics, business management or ecology. Agents typically do this based upon the costs and payoffs for choosing one of either option. This cost-benefit ratio varies depending on the scenario - the game - they are engaged in with other agents.

Some scenarios such as playing chess have very low incentives for cooperation while favoring competition, while others have high incentives toward cooperation. Studying the structure of these games helps us to understand when and why people cooperate.



What is a Game?

In game theory, a game is any context within which adaptive agents interact and in so doing become interdependent. Interdependence means that the values associated with some property of one element become correlated with those of another. In this context, it means that the goal attainment of one agent becomes correlated with the others. This gives us a game, wherein agents have a value system, they can make choices and take actions that affect others and the outcome of those interactions will have a certain payoff for all the agents involved.

The trade negotiations between two nations can be modeled as a game, the interaction of businesses within a market is a game, the different strategies adopted by creatures in an ecosystem can be seen as a game, the interaction between a seller and buyer as they haggle over the price of an item is a form of game. The provision of public goods and the formation of organizations can be seen as games, likewise, the routing of internet traffic and the interaction between financial algorithms are games.

Cooperation vs Competition

To quickly take a simple concrete example of a game, let's think about the current situation with respect to international politics and climate change. In this game, we have all of the world's countries and all countries will benefit from a stable climate, but it requires them to cooperate and all pay the price of reducing emissions in order to achieve this. Although this cooperative outcome would be best for all, it is in fact in the interest of any nation state to defect on their commitments as then they would get the benefit of others reducing their pollution without having to pay the cost of reducing their own emissions.

Because in this game it is in the private interests of each to defect, in the absence of some overall coordination mechanism the best strategy for an agent to adopt given only their own cost-benefit analysis is to defect and thus all will defect and we get the worst outcome for the overall system. This game captures in very simplified terms the core dynamic, between cooperation and competition, that is at the heart of almost all situations of interdependence between adaptive agents.

Non-Cooperative Games

A non-cooperative game is one where an element of competition exists and there are limited mechanisms for creating institutions for cooperation. This may be because of the inherent nature of the game we are playing, that is to say, it is a zero-sum game which is strictly competitive and thus cooperation will add no value.

Noncooperation may be a function of isolation, lack of communication and interaction with which to build up the trust that enables cooperation. We see this within modern societies, as these societies have grown in size they have transited from communal cooperative systems based on the frequent interaction of members to requiring formal third parties to ensure cooperation because of the anonymity and lack of interaction between members of large societies. Likewise, there may simply be a lack of formal institutions to support cooperation between members, an example of this might be what we call a failed state where the government's authority is insufficiently strong to impose sanctions and thus cannot work as the supporting institutional framework for cooperation.

Prisoners Dilemma

The prisoner's dilemma game is a classic two player game that is often used to present the core dilemma at the heart of non-cooperative games. Conceive of two prisoners detained in separate cells, interrogated simultaneously and offered deals in the form of lighter jail sentences for betraying the other criminal. They have the option to "cooperate" with the other prisoner by not telling on them, or "defect" by betraying the other. However, if both players defect, then they will both serve a longer sentence than if neither said anything.

If one cooperates while the other defects they face a high cost thus given the knowledge that the other player's best decision is to "defect" each player improves their own situation by switching from "cooperating" to “defecting". The prisoner's dilemma thus has a single equilibrium: where both players choose to defect. Thus without means to support cooperation both will likely defect resulting in the worst outcome for both.

The Social Dilemma

The prisoner's dilemma is a simple game that illustrates the broader concept of what is call a "social dilemma". Social dilemmas are characterized by two properties: The social payoff to each individual for defecting behavior is higher than the payoff for cooperative behavior, regardless of what the other group members do, yet all individuals in the group receive a lower payoff if all defect than if all cooperate. It is a situation where individual rational behavior leads to a situation where everyone is worse off.

Social dilemmas are of interest to many because they reveal the core tension between the individual and the group that is engendered in situations of cooperation. At their core, social dilemmas are situations in which self-interest is at odds with collective interests and they can be found in many situations of interdependence; from resource management to relationship development, to international politics, public goods provision and business management.

Cooperation

A cooperative game is one in which there can be cooperation between the players and they have the same cost. Cooperative games are an example of non-zero sum games. This is because in cooperative games, either every player wins or loses. Cooperation may be achieved through a number of different possibilities, it may be built into the dynamics of the game as would be the case with a positive-sum game where payoffs are positively correlated. In such a case the innate structure of the game creates an attractor towards cooperation because it is both in the interest of the individuals and the whole organization.

A good example of this are the mutually beneficial gains from trade in goods and services between nations. If businesses or countries can find terms of trade in which both parties benefit then specialization and trade can lead to an overall improvement in the economic welfare of both countries, with both sides seeing it as in their interest to cooperate in this organization, because of the extra value that is being generated.

Achieving Cooperation

In almost every situation of interaction between human beings, social or economic, there is an optimal equilibrium where everyone cooperates and all get a good payoff but there is also often a non-cooperative option that is not so good for all involved but a better option if agents are only self-interested and there is no way to enable cooperation between them. In many ways, we can say it is the job of social and economic institutions to try and provide this infrastructure that enables trust between so that everyone chooses the cooperative outcome and all get the best payoff.

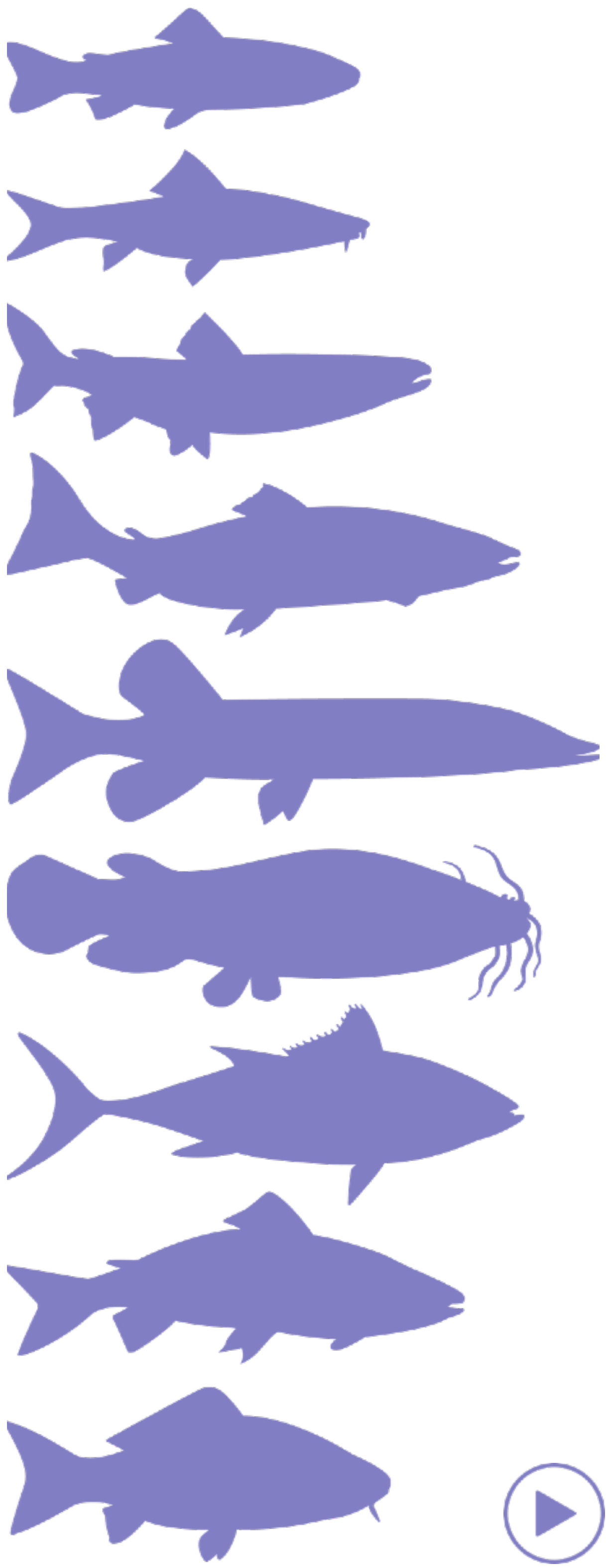
This cooperation may be achieved by external enforcement by some authoritative third party such as governments and contract law. Where we cooperate in a transaction because the third party is ensuring that it is in our interests to do so by creating punishments or rewards. Likewise, cooperation may be achieved through peer-to-peer interaction and feedback mechanisms, as covered in a future guide.

Evolution

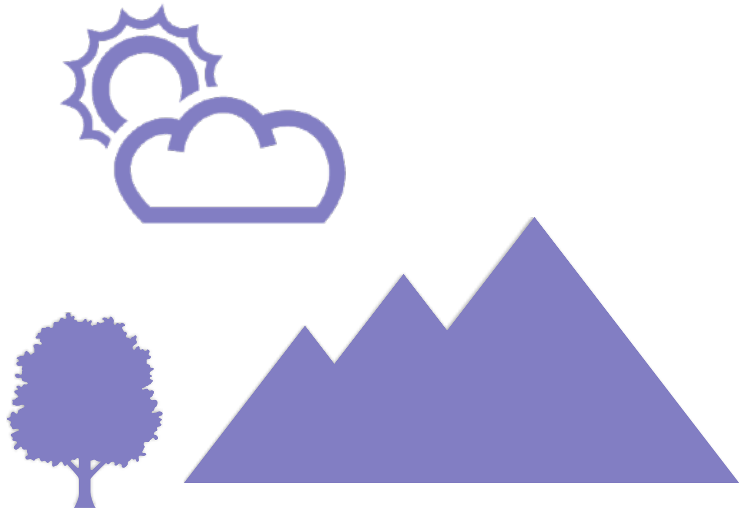
Evolution

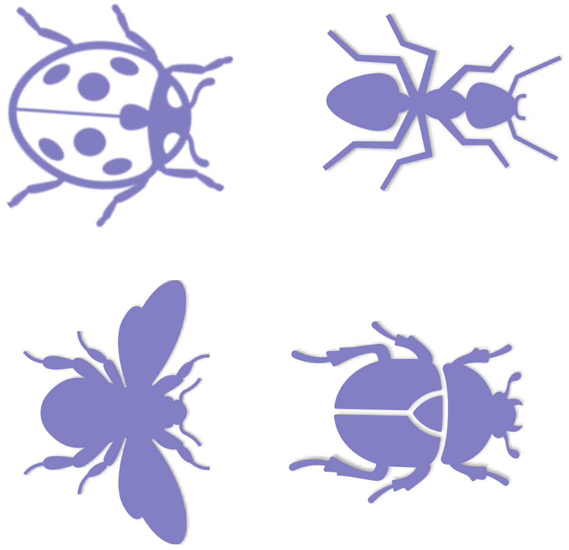
Evolution is a process of adaptation that operates on the macro-level of a system. Adaptation is the capacity to generate a response to some change within the environment. Evolution is this same process but operating on the macro scale, i.e. on the level of a population of agents. Here again, it is the capacity for the system to respond to changes within its environment. Evolution is the adaptive response of a group of entities that occurs over a period of many life cycles. Evolutionary changes reflect the response of the collection of agents to their environment.

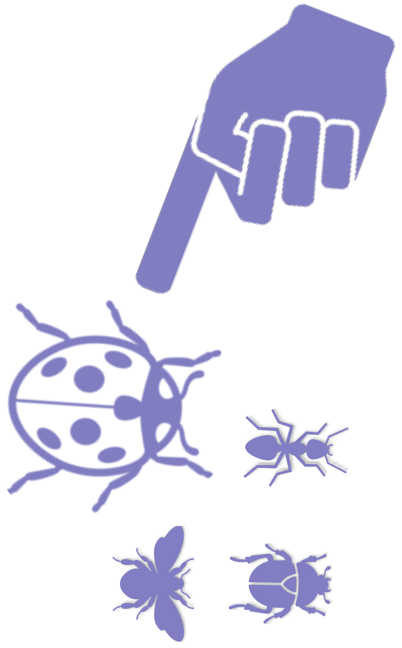
The concept of evolution has a strong association with its application in biology. However, complexity theory deals with the concept on a slightly more abstract level as it applies to all complex adaptive systems from the development of civilization to financial markets, cultures, and technologies. As such, we are trying to understand evolution as a continuous and pervasive phenomenon that occurs in all types of natural, social and engineered systems.

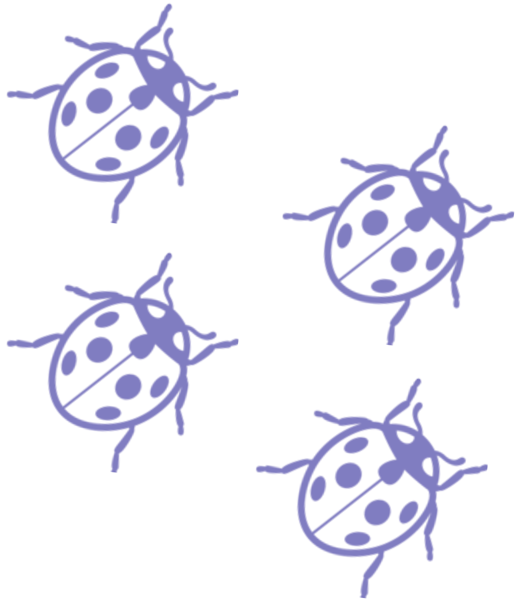


Evolution - How it Works









1. Change

A population of agents exists within some environment and must periodically adapt as a whole in order to survive and maintain functionality.

2. Variety

Responding to changes means selecting from a variety of different internal states or strategies, thus the need for diversity.

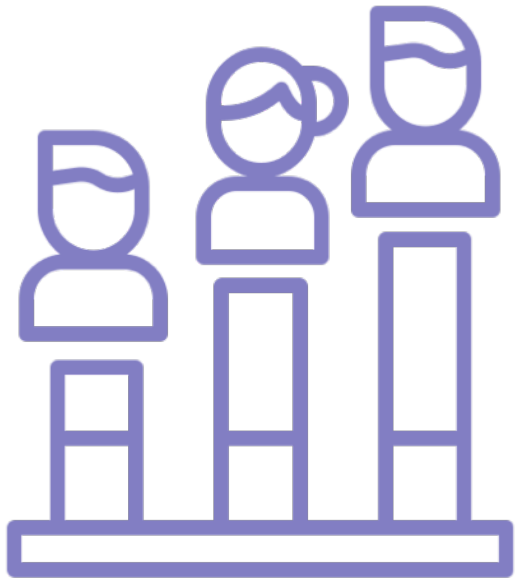
3. Selection

The different variants in the system are exposed to their operating environment to see which are best adapted and most successful within that context.

4. Replication

Those that prove "fittest" attract more resources and are duplicated, thus the system as a whole comes to express more of their characteristics.

Example - Politics

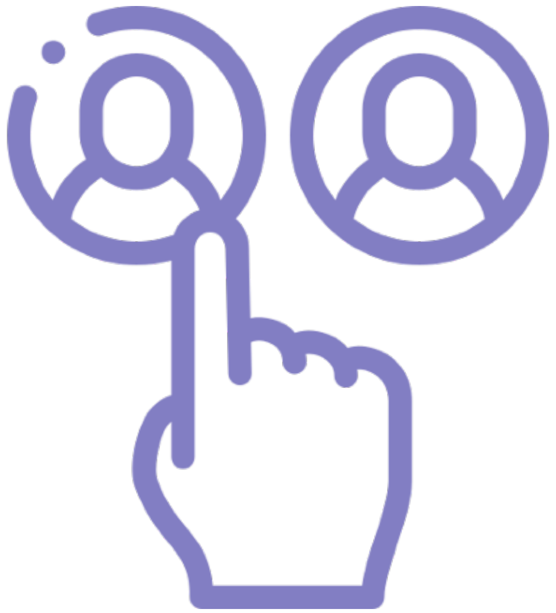


1. Change

Broader changes in culture and society lead to changes in socio-political beliefs.

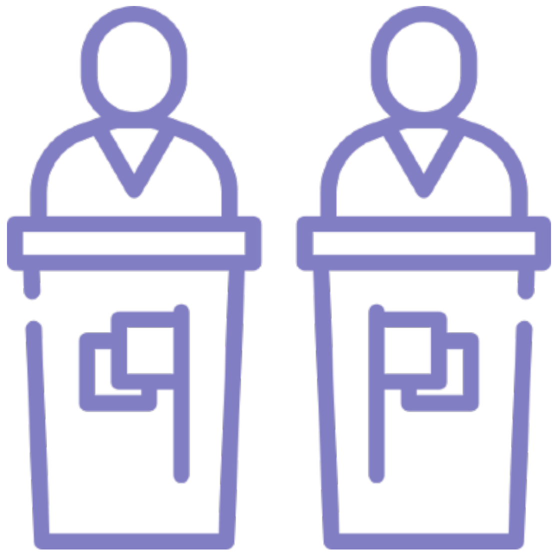
2. Variety

New political parties form to represent the different political perspectives of society



3. Selection

Those which best match the views of the society are selected for at the ballot box.



4. Replication

Those that are selected for become more prevalent in the political system.

Example - Economics

1. Change
Technology changes make possible new products and services.

2. Variety
Businesses create a set of products based upon the new technological possibilities.

3. Selection
Consumers select the product that best suits them in their given context.

4. Replication
Businesses whose products were selected have more revenue to grow while others don't.

Requisite Variety

Through the iterations of evolution over a prolonged period of time, a system can go from starting simple to becoming more complex. By retaining functional variants, it can expand to become capable of operating within broader more complex environments.

The process of evolution illustrates the importance of variety in order to be able to respond to changes within the environment. A system with sufficient internal variety to represent the possible changes within its environment is said to have requisite variety. When a system has requisite variety, then it can be said to have control over itself within that particular environment and be sustainable.

However, there is always a broader environment that will present the system with a wider more complex set of eventualities for it to deal with. Thus evolution proceeds through a process of differentiation - to produce variety - and integration as the system becomes more complex and can operate in a broader more complex environment.

Law of Requisite Variety

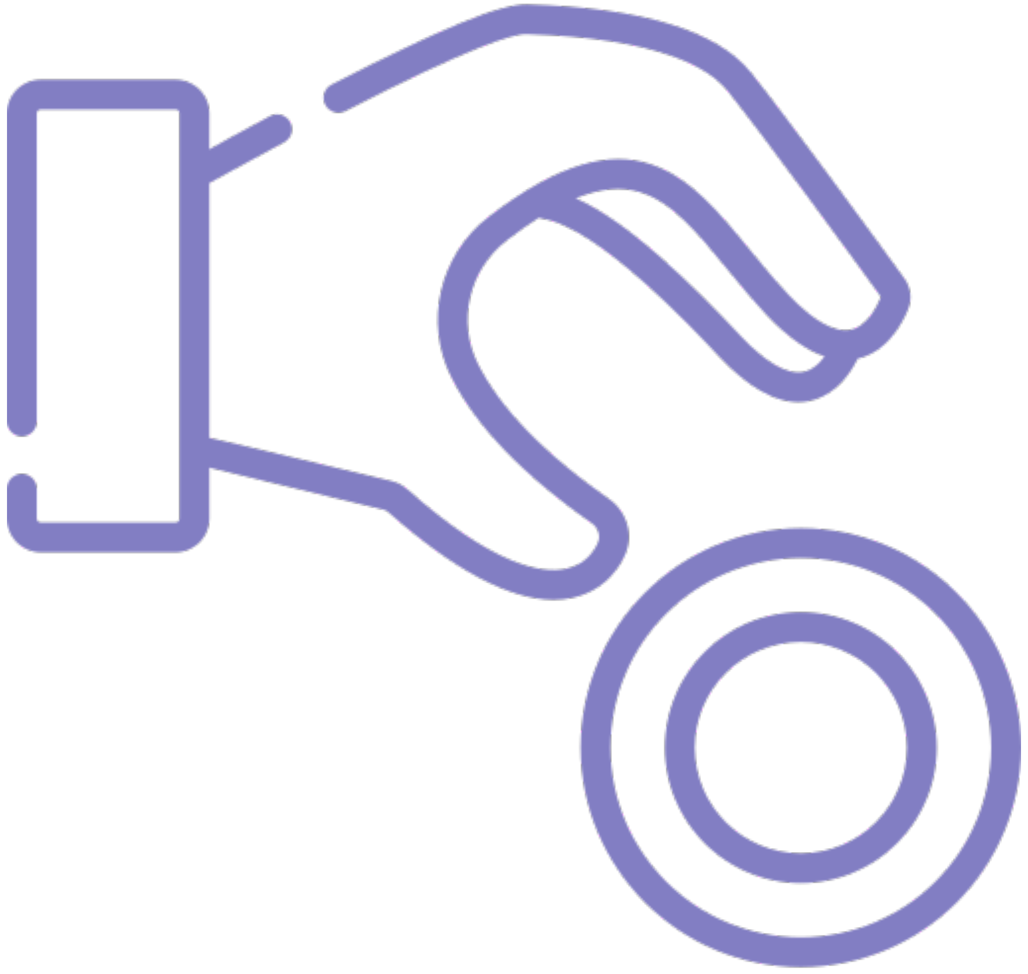
An important theory within cybernetics created by Ross Ashby which states that, if a system is to be stable, the number of states of its regulatory mechanism must be greater or equal to the number of states in the system being regulated. Thus Ashby's Law means that for a system - an organism, a government, a car to be regulated or under control means that for any given change in state presented by its environment the system can generate a response so as to maintain functionality. The system has to be able to respond to the diversity of alterations by choosing from some capacity or response that it has internally to counteract it, thus enabling it to maintain homeostasis and control.

Resilience Theory

Resiliency

Resilience is typically understood as the capacity of a system to maintain functionality given some alteration. When a system is subject to some sort of perturbation or disturbance, it responds by moving away from its initial state. The tendency of a system to remain close to its equilibrium state, despite disturbances, is termed robustness. On the other hand, the speed with which it adapts and finds a new viable state after disturbance may be understood as its resilience.

Whereas resistance will try to remove all disturbances through control and a strong boundary condition, adaptation comes with a recognition of the importance of disturbance in testing the system. This is in order to maintain the competency of the system's components; as the diversity and effectiveness of its constituent elements is the only thing that is going to ensure its preservation. This is illustrated by the development of the human immune system or the resilience of a forest to fires as they both need periodic testing in order to maintain their resilience.



Dependence

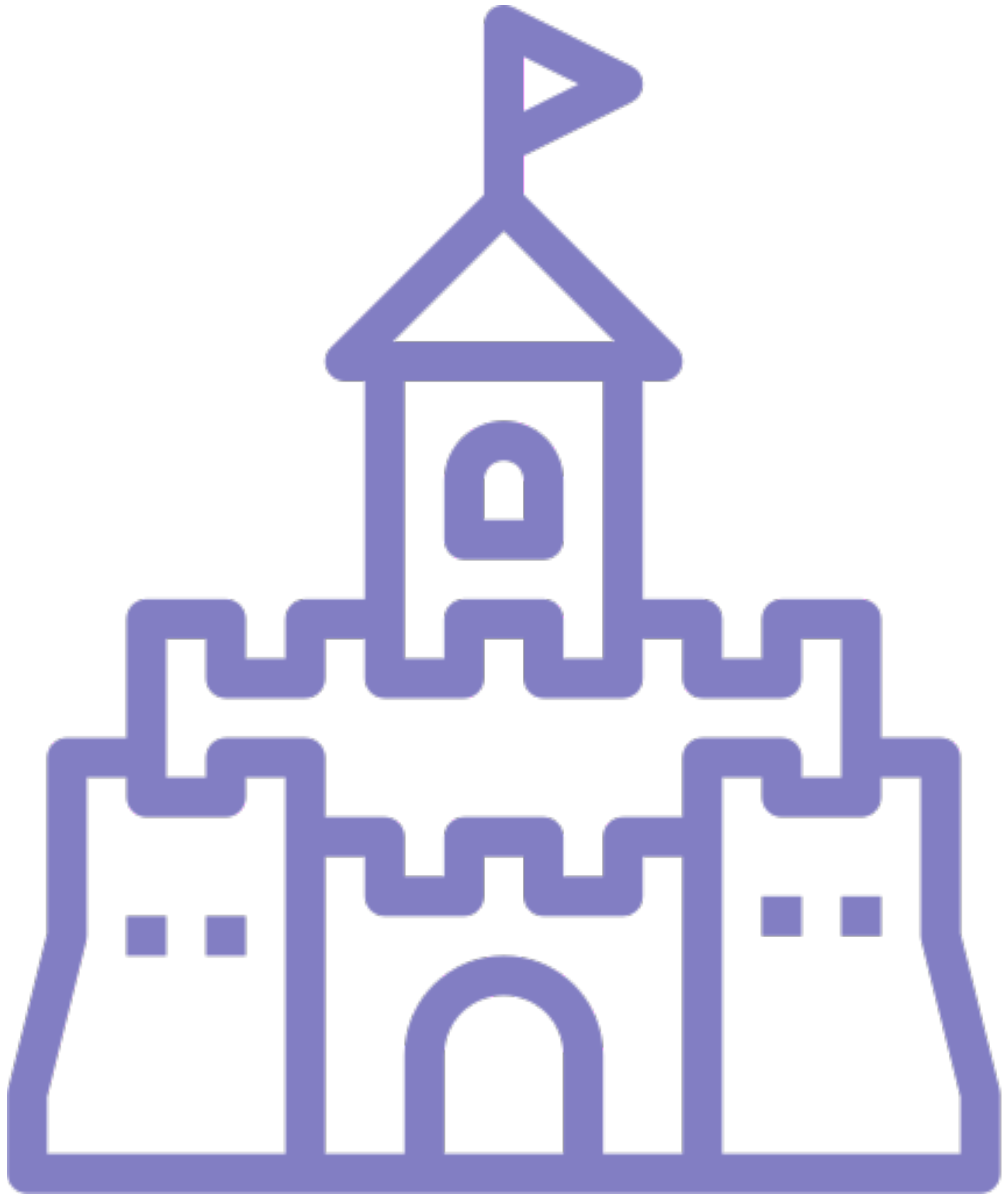
All systems operate within an environment and they are dependent upon some set of input values from that environment in order to operate successfully. Whether this is the human body requiring an input of oxygen or a technology requiring the input of fuel or a government requiring the cooperation of its people. When the system goes outside of this range of required inputs its functionality and structure become degraded. In the way that a human without food will cease to operate effectively or a government without financial revenue will be rendered dysfunctional over time.

Thus in order for the system to maintain functionality it has to maintain itself within a given set of input parameters. There are essentially just two different ways of achieving this. The system can work to control the input values so that they do not change - thus working to resist change - or it can work to enable its capacity to adapt to a wide variety of input values in which case again it would be able to stay within the required input range. This second approach - where the system works to improve its capacity to adapt - we call adaptive capacity.

Antifragile

Resilience is about allowing for changes while looking at the system's ability to endure despite these changes. Whereas resistance and robustness are about achieving "success" - i.e. successfully resisting change is what is valued - with resilience failure becomes more valuable than success as it is only through failing that we build up resilience; resilience is something that is learned through failure. Resilience is always about being aware and ready to adapt to unexpected risks.

The resilience approach is about turning a crisis into an opportunity by seeing it as the potential to develop new capacity. For a resilient community, a change is a learning opportunity, for a fragile community change is a crisis to be avoided. This is the concept of "antifragility" - a term developed by Nassim Nicholas Taleb - which refers to a property of systems that increase in capability to thrive as a result of disturbance, mistakes, shocks, or failures.



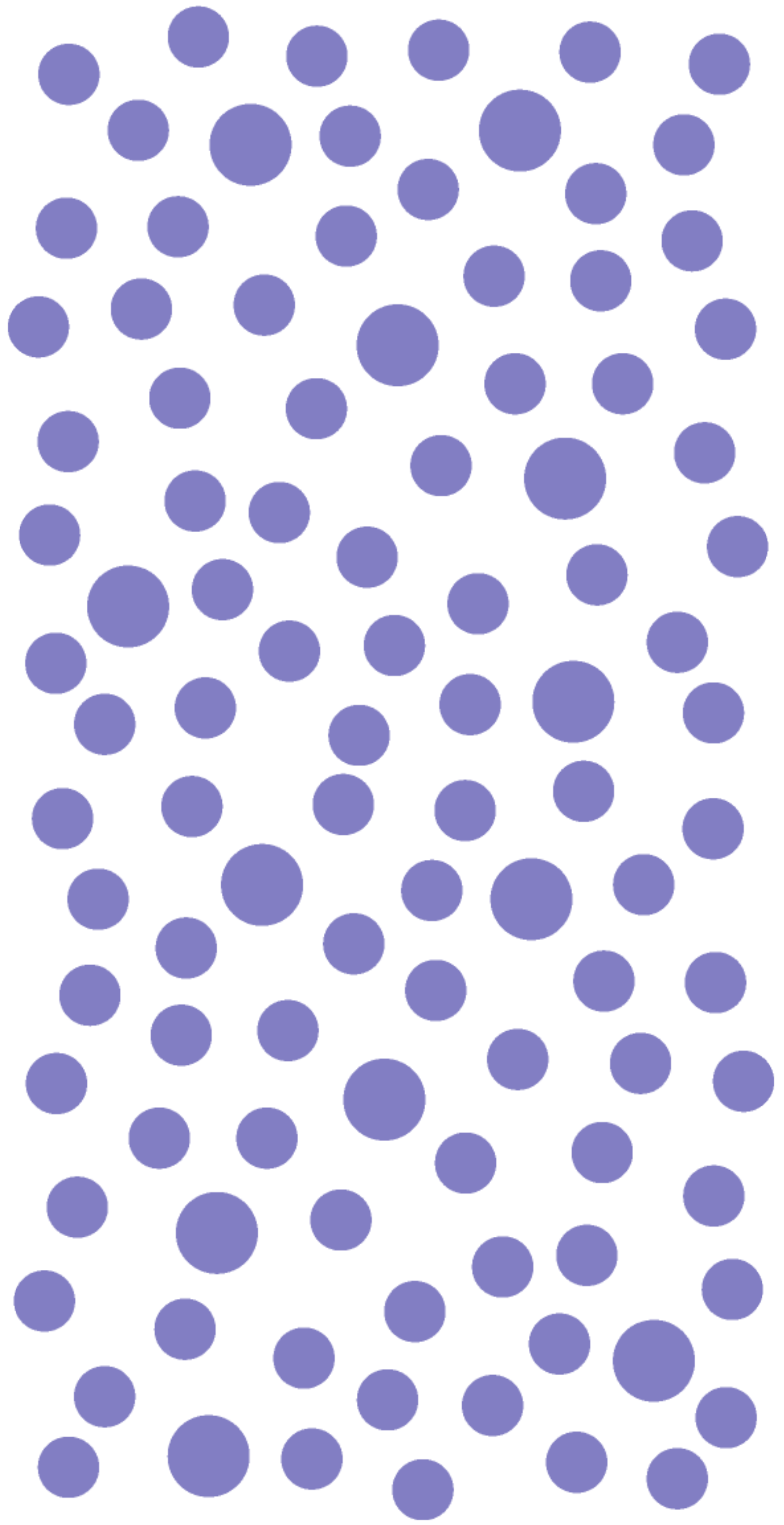
Robustness Strategy

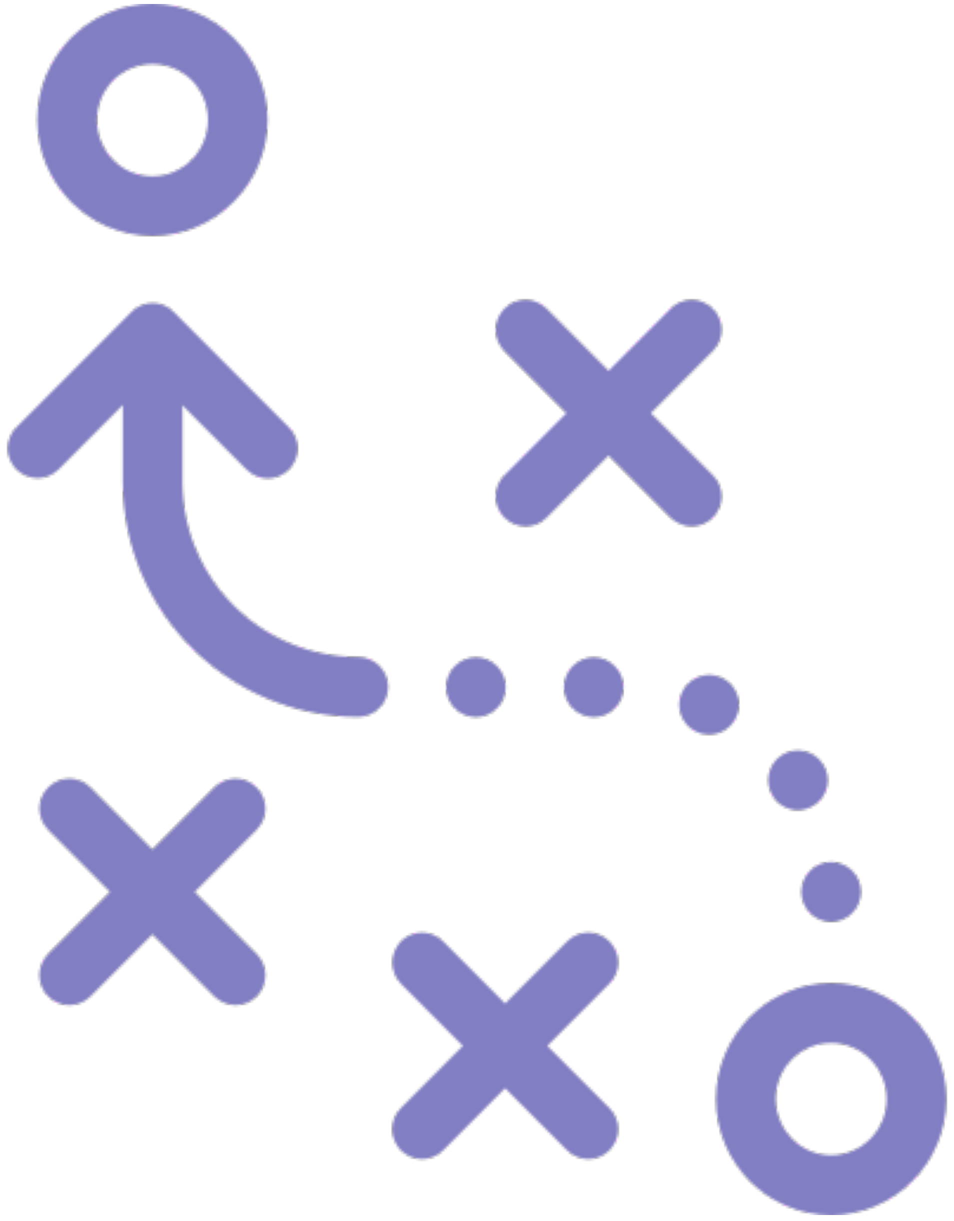
Both robustness and resilience are trying to achieve the same end, of maintaining functionality, but the first does it through resistance while the second does it through adaptation. As always the success of a strategy is contingent upon the environment within which it is being enacted. At low levels of change within relatively stable environments, the robustness strategy works well, but as the change becomes faster and systemic the advantage shifts to the adaptive strategy. We can see this within the business community, as over the past decades with globalization and information technology creating major changes in the business landscape, the terms agility and adaptation have moved to the forefront of management terminology.

Distribution

Robustness and resilience are general characteristics of self-organizing systems both through their capacity to resist change and their capacity to adapt to it. In centralized systems with top-down control, there are specialized components required for regulating the system, these represent largely irreplaceable hubs that will affect the whole system if removed or degraded. Within complex systems in the contrary, control is distributed out on the local level meaning there is much less specialization, missing or damaged components can often be replaced by others and this gives them a much lower level of criticality.

For example, the internet is a distributed system that is not dependent upon one node or route to deliver packages, if a node is damaged or down the computer will send the package through a different pathway. Likewise, self-organizing systems are held within their current configuration by a set of feedback loops that are also distributed out across the system on the local level. A good example of this might be a magnet which consists of many tiny magnetic spins that are all aligned to produce an overall magnetic force, what holds the system together is not one or a few critical parts but distributed feedback relations.





Resiliency Strategy

Resilience becomes of importance when dealing with open systems because, by definition, they involve more exchanges with their environment, while the environment can never be fully controlled; meaning there is a great requirement meant for adaptive capacity and resilience. For example, as societies open up to global processes - as connectivity proliferates and people recognize their interdependence with processes and systems outside of their borders, they must also give up some capacity of control that may have previously existed.

In a globalized world, no nation is able to control the flows of financial capital that affect their economy, no nation is fully able to control the flow of people and goods. In a world where processes that shape people's lives transcends the national level, the strategy of resistance becomes less successful. Resilience in navigating such larger processes of change - as globalization and environmental change - becomes of greater importance to nations and people.

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